Food Coma: Can We Predict It?

A machine learning classification study on post-meal drowsiness (postprandial somnolence) using survey data from 127 respondents — three models, eight features, one question: why do we crash after eating?

Lets_play.html

BI453 · MACHINE LEARNING AND DATA ANALYTICS

By- Akanksha Upadhyay, Sayandeep Majumdar, Vajid Sayyad

Dataset & Problem Setup

The Classification Task

Binary target: Did you feel drowsy after your meal? (Yes = 1, No = 0)

- 84 of 127 respondents (66%) reported drowsiness
- 43 of 127 (34%) did not — a moderate class imbalance
- Handled via class_weight='balanced' across all models

8 Features Selected

- Meal Size, Carb Content, Protein Content (ordinal)
- Stress Level, Meal Heaviness, Sleep Quality (ordinal/numeric)
- Meal Type & Pre-Meal Activity (nominal, one-hot encoded)

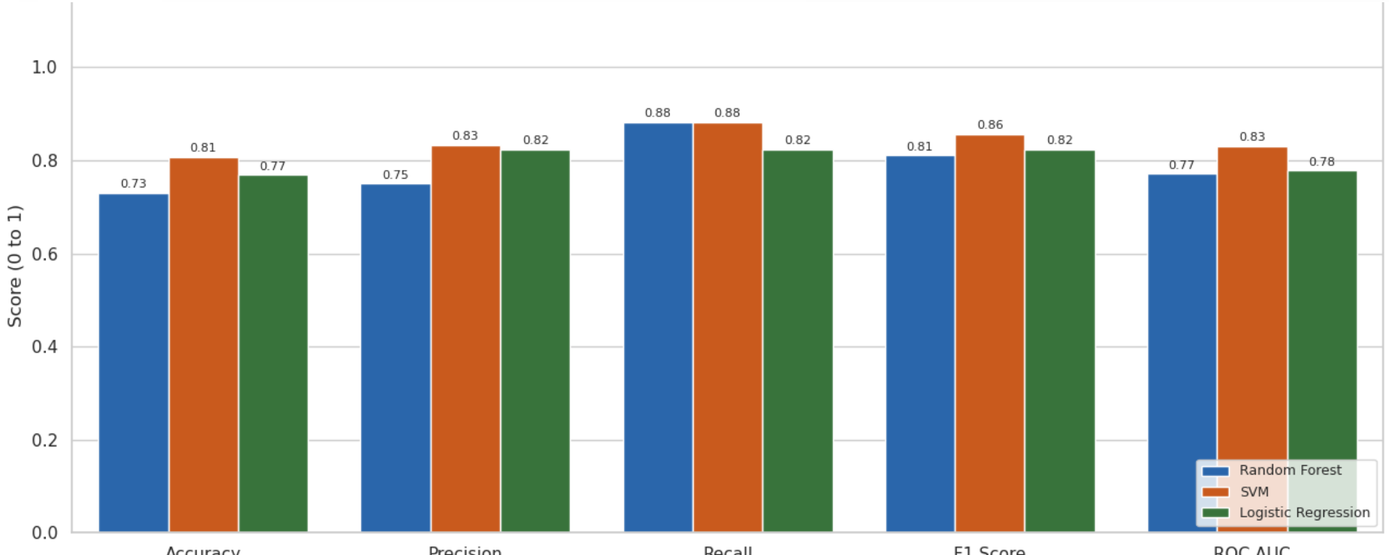
Key Survey Highlights

- Very Large meals → 100% drowsiness rate
- Lunch → highest by meal type (79%); Snacks lowest (33%)
- High Carb Content associated with 77% drowsiness

Model Performance

All three models were wrapped in a scikit-learn Pipeline with a shared ColumnTransformer preprocessor and an 80/20 stratified split (101 train / 26 test samples).

Model	Accuracy	F1 Score	ROC AUC
Random Forest	0.731	0.811	0.771
SVM	0.808	0.857	0.830
Logistic Regression	0.769	0.824	0.778



SVM (Best)

Highest Accuracy, F1 & AUC — maximum-margin hyperplane in high-dimensional space

88% Recall

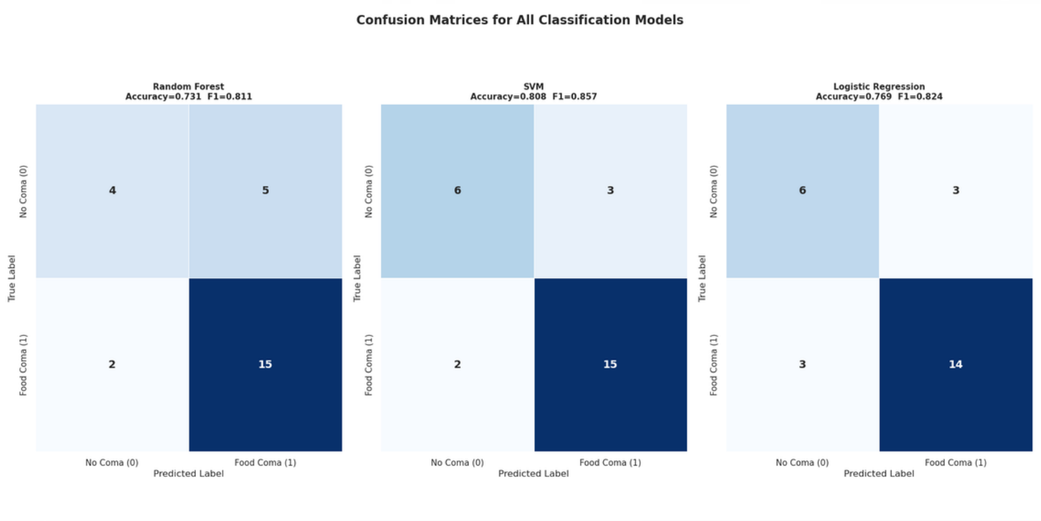
All models caught ~88% of drowsy cases — strong sensitivity across the board

RF Lagged

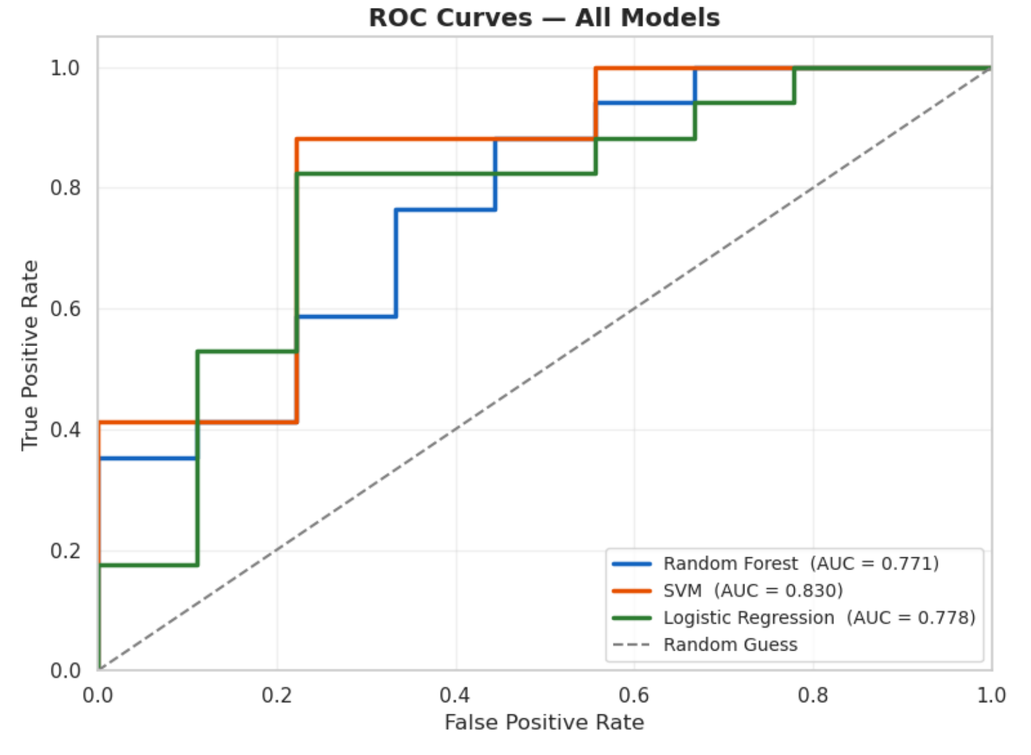
Random Forest struggled most with the non-drowsy minority class despite balanced weighting

Confusion Matrices & ROC Curves

Confusion matrices: SVM and LR correctly classified 6/9 non-drowsy cases; all models predicted 15/17 drowsy cases correctly.

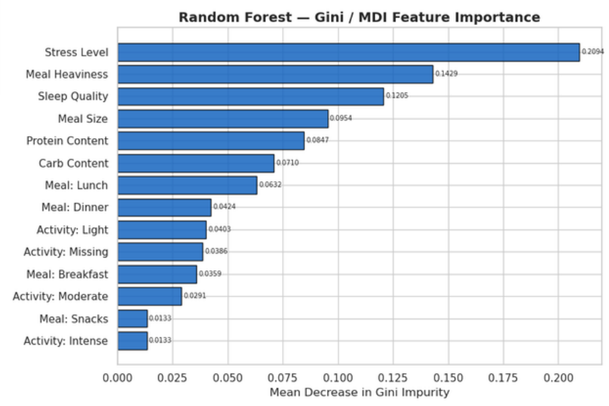


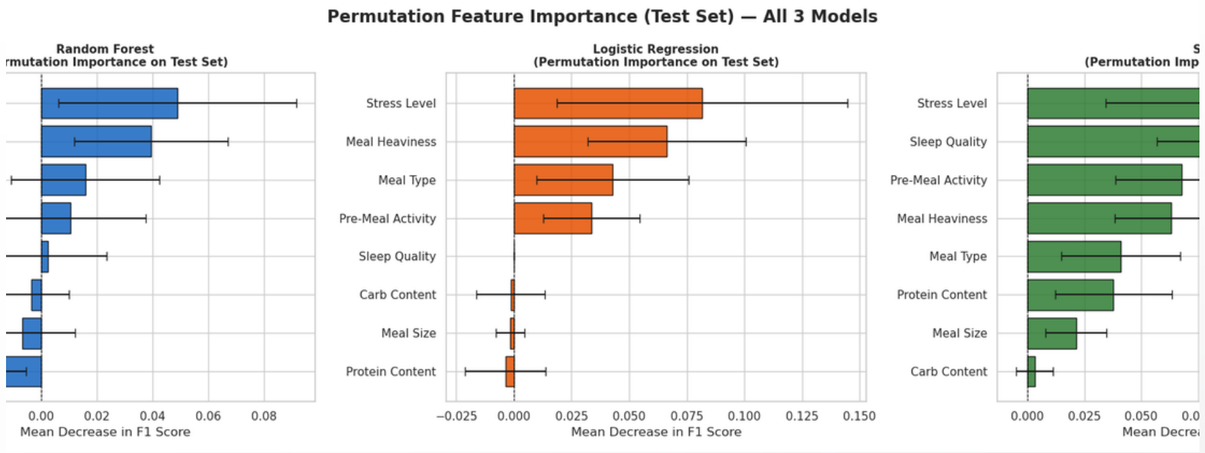
ROC Curves — SVM AUC = 0.830 (best). Random Forest lags, reflecting weaker minority-class separation.



Feature Importance: What Drives Drowsiness?

Three complementary importance methods — Gini/MDI, Permutation, and SHAP — all converge on the same ranking.





Consistent Cross-Method Ranking

1 Stress Level

Gini score 0.2094 — nearly 1.5× Meal Heaviness. SHAP mean ≈ 0.115. Dominant across all three methods.

2 Meal Heaviness

Gini score 0.1429. Consistent second-tier predictor in permutation and SHAP analyses.

3 Sleep Quality

Reliable third-place predictor. Suggests prior sleep deficit amplifies post-meal fatigue.

Meal Size, Carb Content, and Protein Content rank lower — suggesting how you feel before a meal matters more than what you eat.

Feature Importance & Takeaways

Stress Level is the dominant predictor — consistent across Gini/MDI, permutation importance, and SHAP values. Gini score: 0.2094 — nearly 1.5× Meal Heaviness (0.1429). Mean SHAP: ~0.115, far ahead of all other features.

Key Takeaways

Best Model: SVM

Accuracy 0.808 · F1 0.857 · AUC 0.830

Top Predictor: Stress Level

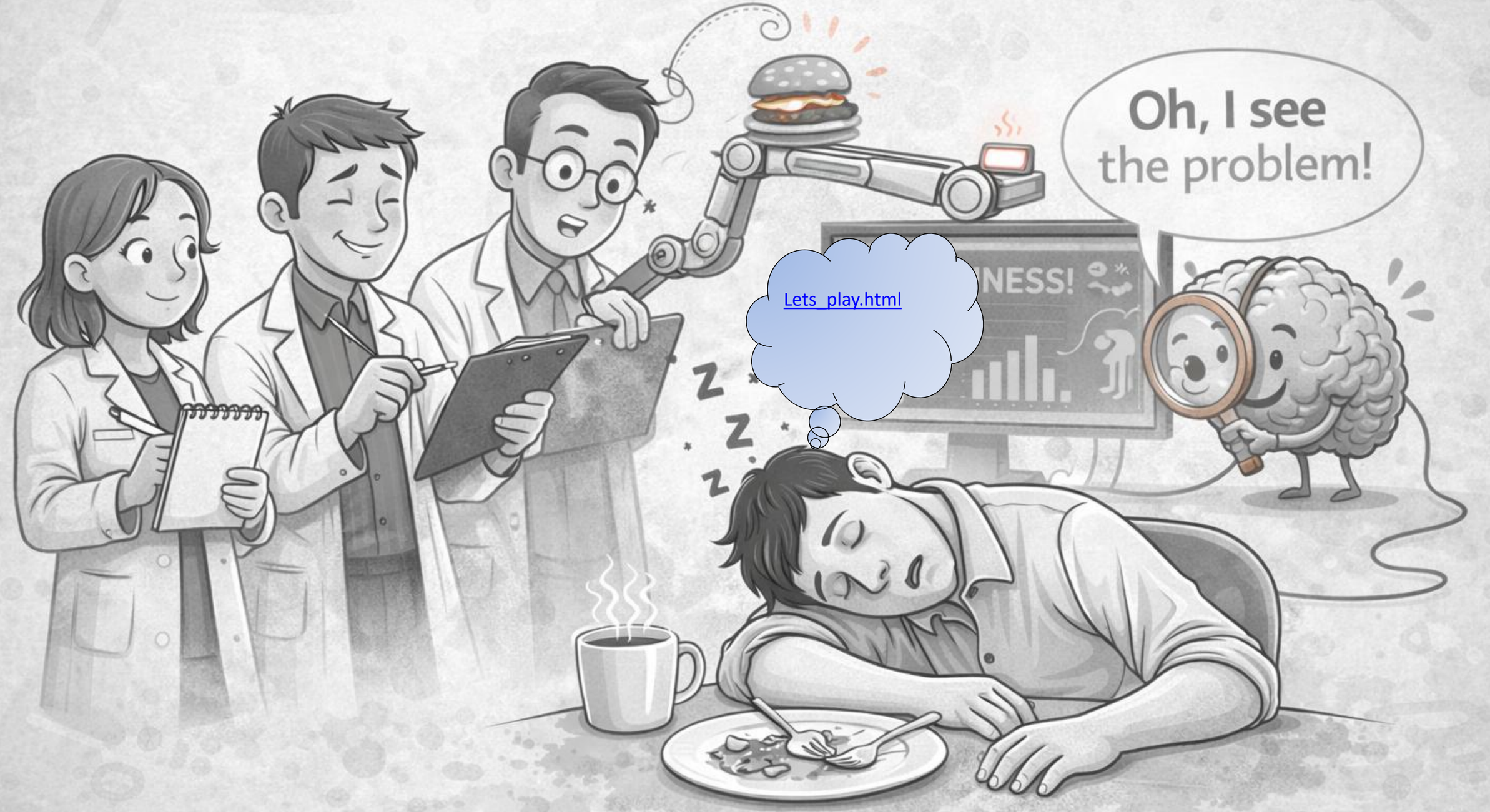
Ranks 1 across all three importance methods. Medium stress – drowsiness

Runner-Up Predictors

Meal Heaviness (2) and Sleep Quality (3) — consistent across models

Interpretation

- High stress → more fatigue
- Heavy meals → digestion → sleepiness
- Poor sleep → increases drowsiness


Oh, I see the problem!
Lets_play.html